

THE MINIMUM RATIO OF TWO EIGENVALUES*

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Abstract. The first two eigenvalues, λ_1 and λ_2 , of the problem $y'' + \lambda\phi(x)y = 0$, $y(\pm\frac{1}{2}) = 0$ are considered. The minimum of their ratio λ_2/λ_1 is sought for $\phi(x)$ ranging over the class of piecewise continuous functions satisfying the inequalities $0 < a \leq \phi(x) \leq A$. It is found that the minimum is an increasing function of a/A , varying from unity at $a/A = 0$ to four at $a/A = 1$. A graph of the minimum is given. The minimizing function $\phi(x)$ is found to be piecewise constant, taking on the value a for $-x_0 < x < x_0$ and the value A elsewhere, and the jump point x_0 is found as a function of a/A . The result provides a lower bound on the ratio λ_2/λ_1 for any $\phi(x)$ in the class considered. The method of analysis is applicable to other similar problems with inequality constraints.

1. Introduction. What is the minimum value of λ_2/λ_1 , where λ_1 and λ_2 are the first and second eigenvalues of the problem

$$(1.1) \quad y''(x) + \lambda\phi(x)y(x) = 0, \quad y(-\tfrac{1}{2}) = y(\tfrac{1}{2}) = 0 \quad ?$$

The minimum is to be found with respect to the coefficient function $\phi(x)$, which ranges over the class of piecewise continuous functions with a bounded number of jumps satisfying the inequalities

$$(1.2) \quad 0 < a \leq \phi(x) \leq A, \quad -\tfrac{1}{2} \leq x \leq \tfrac{1}{2}.$$

We shall solve this problem using a result which is proved by T. Mahar and B. Willner [1]. We shall also consider the minimization of λ_j/λ_k and of other functions of the eigenvalues.

2. Preliminaries. If we replace ϕ by ϕ/A and λ by λA , (1.1) is unchanged and λ_2/λ_1 is unchanged. However ϕ/A satisfies (1.2) with a replaced by $\alpha = a/A$ and with A replaced by 1. Therefore it suffices to seek the minimum of λ_2/λ_1 among $\phi(x)$ in the class C_α of piecewise continuous functions with a bounded number of jumps satisfying

$$(2.1) \quad \alpha \leq \phi(x) \leq 1, \quad -\tfrac{1}{2} \leq x \leq \tfrac{1}{2}.$$

This class of functions is compact. Furthermore λ_1 and λ_2 are continuous functionals of ϕ , and λ_1 is positive and bounded away from zero for $\alpha > 0$. As a consequence, λ_2/λ_1 is a continuous functional of ϕ for ϕ in C_α with $\alpha > 0$, so it takes on both a maximum and a minimum value, we shall denote the minimum by

$$(2.2) \quad \mu(\alpha) = \min_{\phi \in C_\alpha} \frac{\lambda_2(\phi)}{\lambda_1(\phi)}, \quad 0 < \alpha \leq 1.$$

From the definition of the eigenvalues, it follows that $\lambda_2 \geq \lambda_1$, so $\mu(\alpha) \geq 1$.

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Since every eigenvalue of (1.1) is simple, $\lambda_2 > \lambda_1$ and therefore

$$(2.3) \quad 1 < \mu(\alpha), \quad 0 < \alpha \leq 1.$$

Furthermore as α increases, the class C_α decreases and therefore $\mu(\alpha)$ increases or remains constant. Thus

$$(2.4) \quad \mu(\alpha_1) \leq \mu(\alpha_2) \quad \text{if } \alpha_1 \leq \alpha_2.$$

When α has its maximum value, $\alpha = 1$, there is only one function in C_1 , namely $\phi(x) \equiv 1$. For $\phi(x) \equiv 1$ we find that $\lambda_j = (j\pi)^2$, $j = 1, 2, \dots$. Therefore

$$(2.5) \quad \mu(1) = 2^2/1^2 = 4.$$

In summary, $\mu(\alpha)$ exists and is nondecreasing for $0 < \alpha \leq 1$, with $1 < \mu(\alpha) \leq \mu(1) = 4$. All of the preceding remarks apply to $\mu_{jk}(\alpha) = \min [\lambda_j(\phi)/\lambda_k(\phi)]$, $j > k > 1$, except that (2.5) becomes instead $\mu_{jk}(1) = j^2/k^2$. If $j < k$, (2.3) must be replaced by $\mu(\alpha) < 1$. Since $\lambda_k/\lambda_j = (\lambda_j/\lambda_k)^{-1}$, we have $\max_{\phi \in C_\alpha} \lambda_k/\lambda_j = (\min_{\phi \in C_\alpha} \lambda_j/\lambda_k)^{-1} = 1/\mu_{jk}(\alpha)$. Thus our analysis also yields results about the maximum ratio of eigenvalues.

3. Variational analysis. Let us consider a fixed value of α in the interval $0 < \alpha < 1$. Let ϕ_0 denote a function in C_α at which λ_2/λ_1 attains its minimum. We imbed ϕ_0 in a one parameter family of functions $\phi(x, \varepsilon)$ in C_α , with $\phi_0(x) = \phi(x, 0)$, such that the derivative $\phi_\varepsilon(x, 0)$ exists. Then the problem (1.1), with $\phi = \phi(x, \varepsilon)$, has eigenvalues $\lambda_j(\varepsilon)$ and eigenfunctions $y_j(x, \varepsilon)$ which depend upon ε , and are differentiable with respect to ε at $\varepsilon = 0$. Differentiating (1.1) with respect to ε at $\varepsilon = 0$ yields the variational equation

$$(3.1) \quad y_\varepsilon'' + \lambda \phi_0 y_\varepsilon = -\lambda_\varepsilon \phi_0 y - \lambda \phi_\varepsilon y, \quad y_\varepsilon(-\tfrac{1}{2}) = y_\varepsilon(\tfrac{1}{2}) = 0.$$

Here and in the rest of this paper, $\lambda = \lambda(0)$ and $y(x) = y(x, 0)$.

We now multiply (3.1) by y and integrate with respect to x from $-\frac{1}{2}$ to $+\frac{1}{2}$. The left side vanishes because y satisfies (1.1), and we obtain

$$(3.2) \quad 0 = \lambda_\varepsilon \int_{-1/2}^{1/2} \phi_0 y^2 dx + \lambda \int_{-1/2}^{1/2} \phi_\varepsilon y^2 dx.$$

Next we impose upon y the normalization condition

$$(3.3) \quad \int_{-1/2}^{1/2} \phi_0(x) y^2(x) dx = 1.$$

Then (3.2) yields

$$(3.4) \quad \lambda_\varepsilon = -\lambda \int_{-1/2}^{1/2} \phi_\varepsilon(x) y^2(x) dx.$$

Equations (3.2)–(3.4) hold for each eigenvalue λ_j and eigenfunction y_j .

To utilize (3.4) we differentiate λ_2/λ_1 with respect to ε at $\varepsilon = 0$, and replace $\lambda_{1\varepsilon}$ and $\lambda_{2\varepsilon}$ by the expression in (3.4). In this way we get

$$(3.5) \quad (\lambda_2/\lambda_1)_\varepsilon = (\lambda_1 \lambda_{2\varepsilon} - \lambda_2 \lambda_{1\varepsilon}) \lambda_1^{-2} = \frac{\lambda_2}{\lambda_1} \int_{-1/2}^{1/2} \phi_\varepsilon(x) [y_1^2(x) - y_2^2(x)] dx.$$

Since λ_2/λ_1 attains its minimum at $\varepsilon = 0$, the derivative $(\lambda_2/\lambda_1)_\varepsilon$ must be greater than or equal to zero at $\varepsilon = 0$. Therefore (3.5) implies that

$$(3.6) \quad \int_{-1/2}^{1/2} \phi_\varepsilon(x) [y_1^2(x) - y_2^2(x)] dx \geq 0.$$

The inequality (3.6) holds for any admissible variation $\phi_\varepsilon(x)$. At any point where $\phi_0(x) = \alpha$, the admissible variations must satisfy $\phi_\varepsilon(x) \geq 0$. Similarly where $\phi_0(x) = 1$, the admissible variations satisfy $\phi_\varepsilon(x) \leq 0$. However where $\alpha < \phi_0(x) < 1$, the admissible variations can have any value. In this latter case, if ϕ_0 is continuous at x , it follows from (3.6) by a standard argument that

$$(3.7) \quad y_1^2(x) = y_2^2(x), \quad \text{where } \alpha < \phi_0(x) < 1.$$

By the assumed continuity of ϕ_0 at x , (3.7) must hold in an interval containing x . Then by the continuity of y_1 and y_2 , either $y_1 = y_2$ throughout this interval or $y_1 = -y_2$ throughout this interval. In either case, it follows from (1.1) that $\lambda_1 = \lambda_2$. Since this is false, it is not possible for $\phi_0(x)$ to lie in the open interval $\alpha < \phi_0(x) < 1$ at any point where ϕ_0 is continuous. We thus arrive at the following conclusion:

$$(3.8) \quad \phi_0(x) = \alpha \quad \text{or} \quad \phi_0(x) = 1 \quad \text{if } \phi_0 \text{ is continuous at } x.$$

At any point of continuity where $\phi_0(x) = \alpha$, we have seen that $\phi_\varepsilon(x) \geq 0$. Then it follows from (3.6) that $y_1^2(x) - y_2^2(x) \geq 0$. Similarly where $\phi_0(x) = 1$, we have $\phi_\varepsilon(x) \leq 0$ and then (3.6) yields $y_1^2(x) - y_2^2(x) \leq 0$. Then at any point x where ϕ_0 jumps from α to 1 or from 1 to α , it follows by the continuity of y_1 and y_2 that $y_1^2(x) - y_2^2(x) = 0$. Thus we have

$$(3.9) \quad \begin{aligned} |y_1(x)| &\geq |y_2(x)| && \text{if } \phi_0(x) = \alpha, \\ |y_1(x)| &= |y_2(x)| && \text{if } \phi_0 \text{ has a jump discontinuity at } x, \\ |y_1(x)| &\leq |y_2(x)| && \text{if } \phi_0(x) = 1. \end{aligned}$$

All these results, with 1 and 2 replaced by j and k , are valid for the minimization of λ_j/λ_k for any j and k . In particular when we consider the maximum of λ_2/λ_1 over $\phi \in C_\alpha$, the result (3.9) holds with the inequalities reversed. Since $y_1(x)$ is of one sign in the open interval $(-\frac{1}{2}, \frac{1}{2})$, we can choose $y_1(x) \geq 0$ and then replace $|y_1|$ by y_1 in (3.9).

4. Determination of ϕ_0 . In view of (3.8), $\phi_0(x)$ is piecewise constant, taking the values 1 and α in alternate intervals of constancy. Therefore the solution $y_1(x)$ is of the form $N \sin(\lambda_1^{1/2}x + \delta)$, where $\phi_0 = 1$ and of the form $N \sin[(\lambda_1\alpha)^{1/2}x + \delta]$ where $\phi_0 = \alpha$, with N and δ denoting different constants in different intervals. The function $y_2(x)$ is of the same form with λ_1 replaced by λ_2 . The constants N and δ must be chosen to make y_1 and y_2 continuous with continuous first derivatives, to make y_1 and y_2 vanish at $x = \pm\frac{1}{2}$, and to make them satisfy the normalization condition (3.3). For a given ϕ_0 , these conditions determine the constants N and δ up to a common factor of ± 1 multiplying the constants N , and they also determine λ_1 and λ_2 .

The jumps of ϕ_0 must occur at the intersections of the two piecewise sinusoidal curves $y_1(x)$ and $|y_2(x)|$, provided that $y_1(x) \geq 0$. It seems reasonable to conjecture that these curves have exactly two intersections in the open interval $(-\frac{1}{2}, \frac{1}{2})$ and that the intersections are located symmetrically with respect to the midpoint. This is the case when $\alpha = 1$, the intersections occurring at $x = \pm \frac{1}{6}$. This conjecture has been proved by T. Mahar and B. Willner [1]. We shall find the locations of the intersections, the eigenvalues λ_1 and λ_2 and their ratio $\mu(\alpha)$.

We know that $\phi_0(x)$ has two symmetric jumps, say at $\pm x_0$. See Fig. 1. Then $y_1(x)$ is symmetric and $y_2(x)$ is antisymmetric. Thus it suffices to determine these

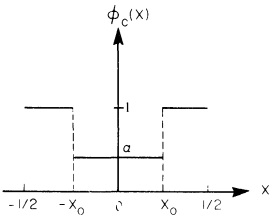


FIG. 1. The function $\phi_0(x)$ which minimizes λ_2/λ_1 in the class C_α of piecewise continuous functions $\phi(x)$ satisfying $\alpha \leq \phi(x) \leq 1$.

functions in the interval $0 \leq x \leq \frac{1}{2}$, requiring that $y_1'(0) = 0$ and $y_2(0) = 0$ because of symmetry and antisymmetry respectively. Since $y_2(0) = 0 < y_1(0)$, it follows from (3.9) that $\phi_0(0) = \alpha$. Therefore we have

$$(4.1) \quad \begin{aligned} \phi_0(x) &= \alpha, & -x_0 < x < x_0, \\ \phi_0(x) &= 1, & |x_0| < |x| \leq \frac{1}{2}. \end{aligned}$$

Now we can write y_1 and y_2 in the forms

$$(4.2) \quad y_1(x) = \begin{cases} N_1 \sin \lambda_1^{1/2}(\frac{1}{2} - x_0) \cos(\lambda_1 \alpha)^{1/2} x, & 0 \leq x \leq x_0, \\ N_1 \cos(\lambda_1 \alpha)^{1/2} x_0 \sin \lambda_1^{1/2}(\frac{1}{2} - x), & x_0 \leq x \leq \frac{1}{2}. \end{cases}$$

$$(4.3) \quad y_2(x) = \begin{cases} N_2 \sin \lambda_2^{1/2}(\frac{1}{2} - x_0) \sin(\lambda_2 \alpha)^{1/2} x, & 0 \leq x \leq x_0, \\ N_2 \sin(\lambda_2 \alpha)^{1/2} x_0 \sin \lambda_2^{1/2}(\frac{1}{2} - x), & x_0 \leq x \leq \frac{1}{2}. \end{cases}$$

These expressions satisfy the boundary conditions $y_1'(0) = y_1(\frac{1}{2}) = y_2(0) = y_2(\frac{1}{2}) = 0$. They also satisfy the differential equation (1.1) with $\phi = \phi_0(x)$, and they are continuous at x_0 .

We must now impose the following five conditions: the normalization (3.3) of y_1 and y_2 , continuity of $y_1'(x)$ and $y_2'(x)$ at x_0 , and the equality $y_1(x_0) = y_2(x_0)$. These five conditions will determine the normalization constants N_1 and N_2 , the eigenvalues λ_1 and λ_2 , and the jump point x_0 . The normalization conditions can be

solved for N_1 and N_2 with the results

$$(4.4) \quad N_1 = \left[\alpha \sin^2 \lambda_1^{1/2} \left(\frac{1}{2} - x_0 \right) \left\{ x_0 + \frac{\sin 2(\lambda_1 \alpha)^{1/2} x_0}{2(\lambda_1 \alpha)^{1/2}} \right\} \right. \\ \left. + \cos^2 (\lambda_1 \alpha)^{1/2} x_0 \left\{ \frac{1}{2} - x_0 - \frac{\sin 2\lambda_1^{1/2} (\frac{1}{2} - x_0)}{2\lambda_1^{1/2}} \right\} \right]^{-1/2},$$

$$(4.5) \quad N_2 = \left[\alpha \sin^2 \lambda_2^{1/2} \left(\frac{1}{2} - x_0 \right) \left\{ x_0 - \frac{\sin 2(\lambda_2 \alpha)^{1/2} x_0}{2(\lambda_2 \alpha)^{1/2}} \right\} \right. \\ \left. + \sin^2 (\lambda_2 \alpha)^{1/2} x_0 \left\{ \frac{1}{2} - x_0 - \frac{\sin 2\lambda_2^{1/2} (\frac{1}{2} - x_0)}{2\lambda_2^{1/2}} \right\} \right]^{-1/2}$$

The remaining three equations are

$$(4.6) \quad \alpha^{1/2} \sin \lambda_1^{1/2} (\tfrac{1}{2} - x_0) \sin (\lambda_1 \alpha)^{1/2} x_0 = \cos (\lambda_1 \alpha)^{1/2} x_0 \cos \lambda_1^{1/2} (\tfrac{1}{2} - x_0),$$

$$(4.7) \quad \alpha^{1/2} \sin \lambda_2^{1/2} (\tfrac{1}{2} - x_0) \cos (\lambda_2 \alpha)^{1/2} x_0 = -\sin (\lambda_2 \alpha)^{1/2} x_0 \cos \lambda_2^{1/2} (\tfrac{1}{2} - x_0),$$

$$(4.8) \quad N_1 \sin \lambda_1^{1/2} (\tfrac{1}{2} - x_0) \cos (\lambda_1 \alpha)^{1/2} x_0 = N \sin \lambda_2^{1/2} (\tfrac{1}{2} - x_0) \sin (\lambda_2 \alpha)^{1/2} x_0.$$

By using (4.4) and (4.5) to eliminate N_1 and N_2 from (4.8), we obtain three equations (4.6)–(4.8) for the determination of λ_1 , λ_2 and x_0 .

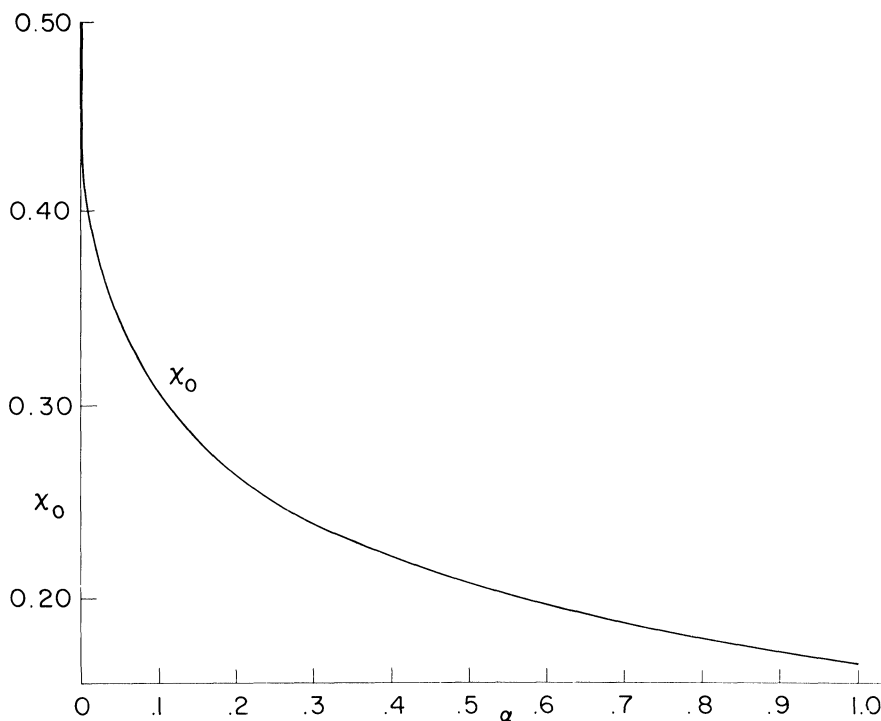


FIG. 2. The point of discontinuity x_0 of the minimizing function $\phi_0(x)$, as a function of α . The value of x_0 is obtained by solving (4.6)–(4.8) numerically for $1 > \alpha > 0.02$, and it is obtained from (4.9) with $c = 1$ for $0.02 \geq \alpha \geq 0$.

We have solved these three equations numerically by Newton's method for various values of α , starting with $\alpha = 1$ and decreasing α by steps of 0.025. For $\alpha = 1$ the solution¹ is $\lambda_1 = \pi^2$, $\lambda_2 = 4\pi^2$ and $x_0 = \frac{1}{6}$. We used this solution as the initial guess for the next smaller value of α , and continued in this way. To proceed to values of $\alpha < 0.075$, we had to decrease the step size to 0.0025, and we could then solve down to $\alpha = 0.02$. The results are shown in Figs. 2 and 3.

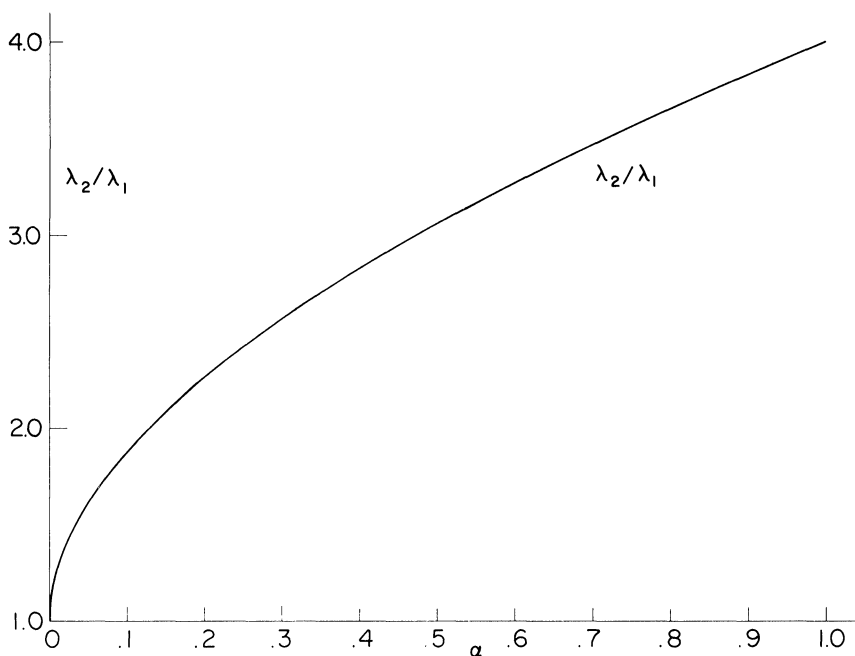


FIG. 3. The minimum value $\mu(\alpha)$ of the ratio λ_2/λ_1 as a function of α . The values λ_1 and λ_2 are obtained by solving (4.6)–(4.8) for each value of α . Newton's method was used for $1 > \alpha > 0.02$, and the asymptotic solution (4.14) was used for $0.02 \cong \alpha \cong 0$.

For smaller values of α we solve (4.6)–(4.8) asymptotically and find

$$(4.9) \quad x_0 \sim \frac{1}{2} - c\alpha^{1/2} - \tilde{c}\alpha + \cdots,$$

$$(4.10) \quad \lambda_1^{1/2} \sim \frac{\pi}{2c\alpha^{1/2}} + d_1 + \cdots,$$

$$(4.11) \quad \lambda_2^{1/2} \sim \frac{\pi}{2c\alpha^{1/2}} + d_2 + \cdots.$$

From (4.10) and (4.11) we obtain

$$(4.12) \quad \mu(\alpha) \sim 1 + \frac{4c}{\pi} (d_2 - d_1) \alpha^{1/2} + \cdots.$$

The coefficient c is a solution of the equation $\cos(\pi/2c) = 1$ so that $c = 1$.

¹ For $\alpha = 1$, $N_1 = N_2 = (\frac{8}{3})^{1/2}$, $y_1(x) = 2^{1/2} \cos \pi x$, $y_2(x) = 2^{1/2} \sin 2\pi x$.

Then

$$(4.13) \quad d_2 - d_1 = \frac{1}{c} \left(\tan \frac{\pi}{4c} + \cot \frac{\pi}{4c} \right) = 2.$$

With these values, (4.12) becomes

$$(4.14) \quad \mu(\alpha) = 1 + \frac{8}{\pi} \alpha^{1/2} + \cdots.$$

The results (4.9) and (4.14) were used in Figs. 2 and 3 for $0 \leq \alpha \leq 0.02$.

5. Generalizations. Suppose we wish to minimize any continuous function $f(\lambda_{i_1}, \dots, \lambda_{i_n})$ of some n eigenvalues, which is bounded below. The minimum $m(a, A)$ exists and is a function of the two constants a and A in (1.2). If f is homogeneous of degree zero, m depends only upon $\alpha = a/A$. In any case m is a nondecreasing function of A and a nonincreasing function of a , and for $a = A$ its value can be found because then $\phi(x) = a$.

At the minimum, the condition $\partial f / \partial \varepsilon \geq 0$ yields, instead of (3.6),

$$(5.1) \quad \int_{-1/2}^{1/2} \phi_\varepsilon(x) [(\partial f / \partial \lambda_{i_1}) \lambda_{i_1} y_{i_1}^2(x) + \cdots + (\partial f / \partial \lambda_{i_n}) \lambda_{i_n} y_{i_n}^2(x)] dx \leq 0.$$

At any point of continuity where $\phi_0(x) = a$ the bracketed sum in the integrand of (5.1) must be ≤ 0 while at a point of continuity where $\phi_0(x) = A$, this sum must be ≥ 0 . If $a < \phi_0(x) < A$ at a point of continuity then the bracketed sum must vanish in an interval around x . We conjecture that the squares of the eigenfunctions are linearly independent when the λ_{i_j} are distinct. Then this sum cannot vanish in any interval provided the derivatives $\partial f / \partial \lambda_i$ do not all vanish there. It then follows that $\phi_0(x)$ cannot lie in the open interval $a < \phi_0(x) < A$ if ϕ_0 is continuous at x . Thus $\phi_0(x)$ is piecewise constant with the value a where the bracketed sum ≤ 0 and the value A where the sum ≥ 0 , while the jumps occur where the sum vanishes. The above conjecture about the linear independence of the squares of the eigenfunctions on the whole interval has been proved by B. Willner [unpublished] but only if $\phi_0(x)$ is in C^{2n} .

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REFERENCE

- [1] T. MAHAR AND B. WILLNER, *An extremal eigenvalue problem*, Comm. Pure Appl. Math., to appear.